

Detailed Project Report

Bhavani River Bridge & Approach Road, Sirumugai · DPR TNHD/CBE/MTP/2025/BR-14 · Highways Department,
Government of Tamil Nadu

1. Public purpose

The Bhavani is crossed at Sirumugai by a low-level causeway which is overtopped during the north-east monsoon. In an average year the crossing is impassable on 24 to 40 days, isolating the villages on the south bank from the taluk headquarters at Mettupalayam, from the Sathyamangalam road, and from the nearest hospital.

This project provides a two-lane high-level bridge of 312 m across the river at the same crossing, with approach roads on both banks totalling approximately 1.6 km. The alignment is set square to the flow so that piers stand with the current rather than broadside to it.

The public purpose is infrastructure of the kind described in s.2(1)(b)(i) of the Act — a public road maintained by the State Government.

2. Land required

60 plots across 3 revenue villages (Kizhakku Sirumugai, Alangombu, Thenkarai Pudur) in Mettupalayam taluk, totalling 16.0445 hectares.

No land is sought between the abutments: the deck spans the river and the bed is not acquired. The land taken is the approach embankment either side, together with the abutment footprints and the temporary construction corridor.

Of the 60 plots, the classification breakdown is: 8 house sites, 44 punjai (dry), 8 nanjai (wet).

3. Alternatives considered

Three alignments were examined. A crossing 900 m upstream would have shortened the span but required acquisition of 4.1 ha of house sites in Kizhakku Sirumugai and the displacement of a further 31 households; it was rejected on that ground.

A crossing 1.4 km downstream avoided all house sites but added 2.7 km of approach road across nanjai land under two paddy crops a year, increasing both the land take and the livelihood impact.

The alignment proposed here takes the least land of the three and displaces the fewest households, at the cost of a marginally longer span.

4. Estimated cost

Civil works Rs. 48,60,00,000; land acquisition and R&R Rs. 21,24,00,000; utility diversion and contingencies Rs. 3,86,00,000. Total Rs. 73,70,00,000.

The land acquisition and R&R estimate is provisional and is superseded by the award once compensation is determined under s.23.